Why Bilinear GCIQL Wins on OGBench Cube and Not Elsewhere: Pre-Registered Phenomena, Mechanisms, and Kill-Tests

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Scope. This note answers the two project questions — **(RQ1)** does the bilinear form work well across environments, and if not, what makes it work so well in OGBench; **(RQ2)** what property is the bilinear form leveraging — by proposing one falsifiable *phenomenon* per question factor, stated mathematically, each with a concrete kill-test implementable in this repo. Every empirical number below is from our own runs (the 288-run MW/MS campaign in `results/table.csv`; the completed cube metric-probe matrix in `/scratch/.../factored-fb-metric-probe/`; dataset audits of the training `npz` files). The companion note `decoding-bilinear.tex` supplies the kernel formalism; this note supplies the causal story and its tests.

1 The evidence to be explained

Notation: critics $\in$ {BIQL (bilinear GCIQL), GCIQL (monolithic), CRL, FB, QRL, VIP}; actors $\in$ {BCFLOW (flow-matching BC, q_{coeff}=0: the critic *never* touches the policy), FLOWBC (identical plus the critic's Q-ascent term), DDPGBC}. All critics share HPs across environments: $\gamma = 0.99$, expectile $\tau = 0.9$, lr $3{\cdot}10^{-4}$, batch 1024, hidden $[512, 512, 512]$ + LayerNorm, goal sampling $(p_{\text{cur}}, p_{\text{traj}}, p_{\text{rand}}) = (0.2, 0.5, 0.3)$ geometric.

E1 Cube (OGBench `cube-single-play-v0`, 500k, 3 seeds). BIQL-d50+FLOWBC $= 92.7 \pm 1.4$ (fresh reproduction of the historical 90.7); BIQL-d512 $= 89.2$; GCIQL$= 73.2 \pm 1.3$; CRL$= 28$–34; pure-BC floor (BCFLOW arms) $= 8$–10. Thus on cube the *entire* signal is critic guidance: +84 points for BIQL, +64 for GCIQL, +20 for CRL over the same BC policy.

E2 Metaworld. The BC floor is already the ceiling: BCFLOW alone scores 0.89–1.00 on play-noise (`mw-push`, `mw-pick-place`, historical `mw-bin-picking`, `mw-button-press`) and 0.58–0.93 on expert. Adding the critic ($\rightarrow$FLOWBC) changes success by -0.27 to $+0.00$; BIQL$\approx$GCIQL everywhere.

E3 ManiSkill (play-noise). Critic guidance has *negative* sign for every goal-conditioned critic and *positive* sign for FB: on `ms-pick-cube`, BCFLOW$\rightarrow$FLOWBC moves BIQL $0.38 \rightarrow 0.08$, GCIQL $0.42 \rightarrow 0.14$, CRL $0.40 \rightarrow 0.10$, but FB $0.57 \rightarrow 0.69$.

E4 The bilinear critic does *not* have the best temporal geometry. Probe P1 (Spearman of $\hat{d}$ with time-gap, partialed on raw-obs distance): GCIQL $0.63/0.54 >$ BIQL $0.54/0.41$. RSA across methods: GCIQL$\leftrightarrow$QRL $= 0.88$, GCIQL$\leftrightarrow$CRL $= 0.54$ (a three-method consensus geometry), while BIQL agrees with nobody (≤ 0.21) and is closest to *raw observation distance* (0.45–0.52). Yet BIQL is the best policy by 20 points.

E5 The bilinear solution is a bounded, norm-concentrated, low-rank object. $\text{corr}(\langle\varphi, \psi\rangle, \cos\angle(\varphi, \psi)) = 0.996$; norm CVs ≤ 0.09 (both towers, both d); effective rank 28/50 and 48/512; d=5 historically collapses to 9.6%. (Caveat: the untrained net already has cos-corr 0.994 — norm concentration is architectural, from LayerNorm; what is *learned* is the direction field.)

E6 Geometry saturates early; success keeps climbing. From 100k to 500k steps, P1-partial moves $0.37 \rightarrow 0.41$ while success moves $65 \rightarrow 93$.

E7 The datasets are not the same kind of object. Cube-play: 10^6 transitions, 1000 trajectories of length 1000, task-agnostic scripted play that relocates the block indefinitely (no terminal dwell). MW: 10^5 transitions, length 100, (noisy-)expert reaching one target then *dwelling at the goal for 48–71% of all recorded steps.* MS: $2.5{\times}10^4$ transitions, length 25, median first-success at step 5 (expert) / 11 (play-noise), dwell 27–75%. "Play-noise" for MW/MS is ε-mixed noisy *expert* ($\varepsilon_{\max} = 0.25$, $\sigma_{\max} = 0.15$, 30% clean) — single-corridor data, not play.

E8 The goal channel is structurally different. MW/MS are goal-*observable*: dims 36:39 (MW) / 26:28 (MS) of the observation are the episode's target coordinates, constant within an episode (std $< 10^{-6}$), redrawn across episodes. OGBench cube observations contain *no* goal information. Consequently on MW/MS: (i) the $p_{\text{rand}} = 0.3$ cross-trajectory relabeled pairs pair a state whose embedded target is c with a goal whose embedded target is $c' \neq c$ — physically unreachable pairs; (ii) at eval, `env/unified.py:167` conditions every episode on a *single fixed dataset frame* (argmax-reward next-obs) while the environment redraws its target every reset (`_freeze_rand_vec=False`); (iii) FB is immune because its $z = \mathbb{E}[r\, B(s')]$ is inferred from the recorded *task* rewards, not from a goal frame.

2 Setup

Let observations decompose as $x = (s, c)$ where s is the controllable state and c an episode-immutable context (the embedded target; on cube, $c = \varnothing$). Let $d^*(x, x')$ be the minimal dataset-supported hitting time and $K^* = \gamma^{d^*} \in [0, 1]$ the discounted reachability kernel (`decoding-bilinear.tex`, Eq. 2). A goal-conditioned critic trained on relabeled pairs estimates $\hat{K} \approx K^*$ on the sampled pair distribution μ. The one-step actor solves

$$\max_{\theta}\ J(\theta) = \mathbb{E}_{s,g}\Big[\, q \cdot \hat{Q}\big(s, \pi_\theta(s, g), g\big)\ -\ \beta\, \|\pi_\theta(s, g) - \pi_{\text{BC}}(s, g)\|^2 \Big], \tag{1}$$

with $q \in \{0, 1\}$ the BCFLOW/FLOWBC switch. Everything below is about when and why the $\hat{Q}$-term in (1) helps.

3 RQ1: why the cube result does not replicate

3.1 Phenomenon I: guidance headroom censoring

Phenomenon 1 (Headroom censoring)**.** *Write S_q for success of the actor trained with critic weight q in* (1)*. Then*

$$S_1\ =\ S_0\ +\ H \cdot \rho, \qquad H := S_{\max} - S_0\ \ (\textit{headroom}), \quad \rho \in [-S_0/H,\, 1]\ \ (\textit{guidance quality}). \tag{2}$$

The benchmark measures $H\rho$*, not* ρ*. On cube $H \approx 0.9$ and the observed $H\rho$ separates critics by 60+ points (E1). On MW $H \leq 0.1$ (E2):* any *critic — perfect or useless — produces $|S_1 - S_0| \leq 0.1$. The MW experiments are censored, not null: they carry almost no information about ρ, hence none about the bilinear form.*

The censoring is caused by the data, not the simulator: noisy-expert corridors (E7) make π_{BC} already near-optimal for the eval task family. Cube-play is the opposite regime — BC on undirected play is at 9%, so *all* performance must come through $\hat{Q}$.

Prediction 1. *Reopen headroom on MW and the ranking must reappear if and only if ρ differs across critics. Implementation (cheap, data-side only): (i) subsample MW play-noise to $\{25\text{k}, 50\text{k}\}$ transitions and/or drop the 30% clean-expert fraction and raise $\varepsilon_{\max}$ at collection; (ii) rerun the $\{$BIQL, GCIQL$\}\times\{$BCFLOW, FLOWBC$\}$ square, 3 seeds. Falsifier of Ph. 1: BC floor drops (headroom opens) but all $S_1 - S_0$ remain ≈ 0 — then the failure is not censoring but guidance invalidity (Ph. 2).*

3.2 Phenomenon II: contextual goal misspecification

Phenomenon 2 (Ill-posed goal channel on goal-observable benchmarks)**.** *With $x = (s, c)$ and context immutable within an episode, the true kernel factorizes as*

$$d^*\big((s, c), (g, c')\big) = \begin{cases} d^*_c(s, g) & c' = c \\ \infty & c' \neq c \end{cases} \qquad \Longrightarrow \qquad K^*\big((s, c), (g, c')\big) = \mathbf{1}[c{=}c']\, \gamma^{d^*_c(s,g)}. \tag{3}$$

Under relabeling with $p_{\text{rand}} = 0.3$, 30% of value-training pairs sit in the $c \neq c'$ block, whose regression target is the constant $-1/(1-\gamma)$; the critic must spend capacity learning the diagonal-block indicator instead

of geometry. At eval the agent is conditioned on one fixed frame $g_0 = (\bar{g}, c_0)$ while the environment draws $c \sim P(c)$ fresh each episode, so almost surely $c \neq c_0$ and the conditioned pair lies in the unreachable block: the ascent direction $\nabla_a \hat{Q}\big((s,c),a,(\bar{g},c_0)\big)$ is a gradient of a function whose true value is constant, i.e. pure critic-error. Its expected contribution to (1) *is adversarial noise on top of BC, so $\rho < 0$: guidance* hurts*, exactly the sign observed on MS (E3), while pure BC survives by reading the true target from its own observation.* FB*'s task-inferred z never touches this channel — explaining its immunity (E3) — so the MW/MS tables compare a well-posed conditioning (*FB*) against an ill-posed one (all GC critics); they are not a test of the bilinear form.*

Prediction 2. *Three independent repairs, each implementable in a day, each with a pre-registered sign: (a)* ***mask the context dims*** *out of the goal input ($\psi(g)$ and actor conditioning) — removes the c-block structure from $\hat{K}$; (b)* ***per-episode goal synthesis at eval****: build g from the* current *episode's observation by writing the target coordinates into the object slots (both MW and MS expose them), instead of the fixed dataset frame; (c) $p_{\text{rand}} = 0$ for MW/MS — removes the unreachable block from training. Ph. 2 predicts (b) alone flips the guidance sign on* `ms-pick-cube` *($S_1 - S_0$ from -0.30 toward ≥ 0), and (a)+(c) increase it further. Falsifier: repairs applied, headroom present (MS has $H \approx 0.6$), and GC guidance still negative — then the defect is in the value target itself (Ph. 3).*

3.3 Phenomenon III: kernel degeneracy under short-horizon dwell data

Phenomenon 3 (Range compression and goal-marginal collapse)**.** *The information the critic can add over BC lives in the* variation *of K^* across the sampled pairs. On trajectories of length H_{traj}, relabeled hitting times satisfy $d^* \leq H_{\text{traj}}$, so*

$$K^* \in [\gamma^{H_{\text{traj}}}, 1]: \qquad \underbrace{1-\gamma^{1000} \approx 1.0}_{cube} \qquad \underbrace{1-\gamma^{100} = 0.63}_{MW} \qquad \underbrace{1-\gamma^{25} = 0.22}_{MS}. \tag{4}$$

On MS, additionally, $(1-\gamma)d^* \leq 0.25$ puts K^* in its linear regime, $K^* \approx 1-(1-\gamma)d^*$, so the kernel matrix is affine in d^* ($K \approx \mathbf{1}\mathbf{1}^\top - (1-\gamma)D$): the exponential structure whose low-rank factorization is BIQL*'s inductive bias (*`decoding-bilinear.tex`*, §2) degenerates, and monolithic and bilinear parametrizations face the same easy regression squeezed into a 0.22-wide range against TD noise. Dwell (E7) compounds this: 48–75% of relabeled goals collapse into the tiny at-goal set, so the goal marginal is nearly a point mass and most pairs have $d^* \approx$ const. In this regime no critic can differentiate actions ($\rho \approx 0$ for all), and the bilinear form has nothing to leverage —* independently *of Ph. 2.*

Prediction 3. *(a)* ***HP-side*** *(cheapest possible test, pure config): set the discount per-environment by the scaling rule $\gamma_{\text{env}} = \exp(-1/\bar{d})$ with $\bar{d}$ the median relabeled hitting time ($\bar{d} \approx 10$ on MS $\Rightarrow \gamma \approx 0.9$; $\bar{d} \approx 30$ on MW $\Rightarrow \gamma \approx 0.97$), restoring the cube-like ratio K-range ≈ 1. Prediction: GC critic guidance on repaired-goal MS (after Pred. 2) becomes positive and* BIQL *separates from* GCIQL*. (b)* ***Data-side*** *(the decisive test): collect OGBench-style* play *on MS/MW — long horizon (≥ 500 steps), scripted policy that chains random object relocations, no dwell, no per-episode target reset if the backend allows freezing (*`_freeze_rand_vec=True` *matched between collection and eval). This equalizes E7 across benchmarks. Prediction: with E7 and E8 equalized, the cube ordering (*BIQL $>$ GCIQL $\gg$ BC*) reappears on MW/MS. This is the single experiment that most directly answers RQ1.*

4 RQ2: what the bilinear form is actually leveraging

4.1 Phenomenon IV: the advantage is not a better metric (established negative)

Phenomenon 4 (Metric fidelity does not explain guidance value)**.** *If* BIQL*'s edge came from a more faithful temporal geometry, it would beat* GCIQL *on the geometry probes and agree with the architecturally-constrained reference (*QRL*). The data show the opposite (E4):* GCIQL *tracks temporal distance better, joins the* QRL/CRL *consensus (RSA* 0.88/0.54*), and* loses by 20 points*;* BIQL *is a geometric outlier nearest raw-obs distance and wins. Moreover geometry saturates at 100k while success keeps rising (E6). Hence value-geometry fidelity on the data distribution is neither necessary nor sufficient for guidance value. Whatever the bilinear form buys, it buys it* at the actor interface*, in properties of $\hat{Q}$ invisible to (s,g)-geometry probes.*

This also disposes of the naive reading of H-sim: yes, the critic is numerically a scaled cosine (E5), but so is the untrained net; cosine-ness per se is architectural, not the learned asset.

4.2 Phenomenon V: bounded low-rank guidance (the proposed mechanism)

Phenomenon 5 (Bounded, goal-uniform, low-rank action guidance)**.** *Decompose the actor's gradient signal. For the bilinear critic* $\hat{Q}(s,a,g) = \langle \varphi(s,a), \psi(g) \rangle / \sqrt{d}$ *with norm concentration* $\|\varphi\| \approx \kappa_\varphi$, $\|\psi\| \approx \kappa_\psi$ *(E5:* $CV \leq 0.09$*):*

$$\textit{(boundedness)} \qquad |\hat{Q}| \;\leq\; \kappa_\varphi \kappa_\psi / \sqrt{d} \;=:\; \kappa \qquad \textit{uniformly in } (s,a,g), \tag{5}$$

$$\textit{(goal-uniform scale)} \qquad \nabla_a \hat{Q} \;=\; J_a \varphi(s,a)^\top \psi(g)/\sqrt{d}, \quad \|\nabla_a \hat{Q}\| \leq \|J_a \varphi\|_{\mathrm{op}}\, \kappa_\psi / \sqrt{d}, \tag{6}$$

$$\textit{(rank restriction)} \qquad \nabla_a \hat{Q} \in \mathrm{row}(J_a \varphi) \textit{ with } \hat{Q} \textit{ of rank} \leq d \textit{ (eff. rank } \approx 28).$$

Consequences, each targeting a known failure mode of critic-guided actors:

(a) ***Exploitation bound.*** *The actor ascends* $\hat{Q}$*, and its true regret relative to BC obeys the standard decomposition* $J^\pi \geq J^{\pi_{\mathrm{BC}}} + \Delta_{\hat{Q}} - 2\sup_a |\hat{Q} - Q^*|$ *over reached actions. For a monolithic MLP the sup is uncontrolled off-data — the actor manufactures phantom value — while* (5) *caps what ascent can ever collect at* 2κ*. The observed* GCIQL *triangle-violation* magnitude *of* 9.5 *steps versus* BIQL*'s* 4.8 *(at comparable rates) is exactly the signature: monolithic errors are rarer but huge; bilinear errors are small because the hypothesis class cannot express large outliers around a rank-28 solution.*

(b) ***Goal-uniform guidance scale.*** FLOWBC *normalizes the Q-term by the* global *mean* $|\hat{Q}|$*; per-goal scale heterogeneity survives that normalization. By* (6) *with* κ_ψ *concentrated (CV* 0.07*), the bilinear critic feeds every goal's gradient at the same magnitude, so no goal region dominates the actor's batch gradient. The monolithic critic offers no such guarantee.*

(c) ***Matrix completion to novel eval pairs.*** *Eval pairs* (s, g^{task}) *are combinations never co-observed in play data. A rank-d factorization is determined by marginal coverage — each state seen with* some *goals, each goal with* some *states — and fills in unseen blocks of* K^* *(matrix-completion generalization), which is precisely where an unstructured MLP must extrapolate freely. This is why* BIQL *can look* worse *on-distribution (E4 probes are val-set pairs) yet act better on eval pairs.*

Finally, the mechanism is a ***conjunction****, which our own 500k matrix already factorizes (E1): the guidance target must be in-sample* optimal *(expectile), and the parametrization must be bounded-low-rank:*

	monolithic	*bilinear*
behavior-policy target (NCE)	—	CRL: 28–34%
in-sample-optimal target (τ=0.9*)*	GCIQL: 73%	BIQL: **93**%

(BC floor 9%*.)* CRL *shares the bilinear geometry and norm concentration (E5 holds for it too) but ranks actions by* behavior *occupancy, so its guidance is capped;* GCIQL *shares the optimal target but not the bounded-low-rank guidance. Only the conjunction clears* 90%*.*

Prediction 4 (Mechanism kill-tests, all on cube, all single-file changes)**.** *Each sub-claim gets its own control; together they separate (a), (b), (c):*

1. ***Bounded monolithic*** *(tests (a) alone): train* GCIQL *with output squashing* $\hat{Q} = \kappa \tanh(\mathrm{MLP}/\kappa)$*,* κ *set to the observed bilinear* $\kappa \approx |V|_{\max}$*, or spectral-norm the last layer. Prediction: recovers a large fraction of the* $73 \to 93$ *gap. If it recovers* none*, (a) is falsified and (c) carries the phenomenon.*
2. ***De-concentrated bilinear*** *(tests (a)+(b) from the other side): remove LayerNorm from the* ψ *tower / add a norm-spread penalty so* $CV > 0.5$*. Prediction: success drops toward* GCIQL *despite identical rank.*
3. ***Eval-pair value probe*** *(tests (c), pure analysis on existing checkpoints): run probe P2 (goal-approach monotonicity) along* eval-task *rollouts — the deferred probe — for* BIQL *vs* GCIQL*. Prediction:* GCIQL*'s ranking quality degrades from val-pairs to eval-pairs strictly more than* BIQL*'s (an interaction effect, not a main effect).*

4. ***Rank threshold*** *(tests (c)):* $d \in \{5, 10, 20, 30, 50\}$*, 3 seeds. Prediction: success transitions sharply at* $d \approx$ *effective rank of the solution (*≈ 28*), i.e. between* d=20 *and* d=30*, not gradually.*

5. ***Fourth cell of the*** 2×2*: give* CRL*'s bilinear critic an in-sample-optimal target (expectile-weighted NCE or TD-InfoNCE). Prediction: it joins* BIQL *near* 90%*; conversely* BIQL *with* $\tau = 0.5$ *(SARSA-like) should fall toward* CRL*.*

Remark 1 (Post-registration result: the d-sweep (P4.4) has run)**.** *3 seeds × 500k on cube (*`/scratch/.../factored-fb-metric-probe`*): d=5:* 0.06 ± 0.05*; d=10:* 0.62 ± 0.07*; d=20:* $\mathbf{0.92 \pm 0.02}$*; d=30:* 0.90 ± 0.02*; d=50:* 0.93 ± 0.02*. The transition is sharp —* +56 *points from d=5 to d=10,* +30 *from d=10 to d=20, then flat — but it sits* below *the d=50 solution's effective rank of* 28*: d=20 is already statistically indistinguishable from d=50. Verdict on Pred. 4.4: the strong reading ("success transitions at $d \approx$ eff-rank of the unconstrained solution") is* ***falsified****; the weak reading survives and sharpens: the guidance signal is a low-rank object of intrinsic dimension* $\approx$ 10–20*, and the eff-rank* 28 *measured at d=50* overshoots *the minimal sufficient rank rather than marking a threshold. Two follow-ups: (i) run the probe battery on the $d \in \{10, 20, 30\}$ checkpoints — if learned eff-rank saturates near* 15–20 *regardless of d, the solution rank (not the container size) is the invariant; (ii) d=15 fills the one remaining gap in the curve.*

Follow-up (i) has now run (3 seeds each): learned eff-rank (phi/psi means) is 4.1/8.0/14.7/19.6/28.0 *at d =* 5/10/20/30/50 *(and* 48 *at* 512*) — it does* not *saturate at an invariant solution rank; it tracks the container sublinearly (compression ratio* $0.82 \to 0.56 \to 0.09$ *as d grows). So eff-rank measures container-conditioned spread, not the signal's intrinsic dimension; the only d-invariant quantity in the sweep is the* success saturation point $d^\dagger \in [10, 20]$*. Rank is a capacity budget, not a learned constant.*

Remark 2 (Post-registration result: P4.3 has run — the interaction is confirmed, in a sharper form)**.** *Full* 2×2 *(scoring critic × rollout-generating policy), 3 seeds, 50 temperature-0 eval episodes per run, per-episode Spearman* $\rho(\hat{V}(s_t, g), t)$ *(median over episodes, mean±std over seeds; bump rate =* $\Pr[\hat{V}_{t+1} < \hat{V}_t]$*):*

scored on ↓ / rollouts from	BIQL *policy (*0.95 *succ.)*	GCIQL *policy (*0.73 *succ.)*
BIQL *critic*	0.983 ± 0.003 *(bump* 0.17*)*	0.871 ± 0.060 *(bump* 0.27*)*
GCIQL *critic*	$\mathbf{0.355 \pm 0.239}$ *(bump* 0.38*)*	0.883 ± 0.015 *(bump* 0.27*)*

Pre-registered prediction (gciql degrades from val-pairs to eval-pairs more than biql): ***confirmed*** *— on val-set pairs* GCIQL *ranks temporal distance* better *(P1:* 0.63 *vs* 0.54*), on eval-task rollouts of the successful policy its value signal is noise (*0.355 ± 0.239*, with per-task medians as low as* -0.80*) while* BIQL *is a near-perfect monotone ramp (*0.983*). The sharper form is the* transfer asymmetry*: each critic is coherent along its own policy's trajectories (diagonal* 0.98/0.88*), but only the bilinear critic transfers to the* other *policy's state distribution (*0.87 *vs* 0.36*). The monolithic critic's ranking is intact exactly up to the states its own (worse) policy reaches and breaks on the states the better policy actually visits — which is what Ph. 5(c) (completion to unseen (s, g) blocks) predicts, and what explains how ascent on* $\hat{Q}_{\text{BIQL}}$ *can push the policy* beyond *the critic-fidelity frontier of* $\hat{Q}_{\text{GCIQL}}$ *(*$73 \to 93$*).*

Remark 3 (Post-registration result: P1 has run — first cross-env replication of the ordering (2026-08-03))**.** *Hard-noise MW data (*$\varepsilon_{\max}$=0.5*,* $\sigma_{\max}$=0.3*, no clean episodes; collected dataset success* 0.50/0.54*, dwell* 0.21/0.25*) opened the headroom as predicted: BC floors dropped from* 0.89–1.0 *to* 0.52–0.67*. On* `mw-push` *the cube structure* ***reappears****:* BIQL $0.600 \to 0.867$ *(*$S_1 - S_0 = +0.27$*) vs.* GCIQL $0.667 \to 0.717$ *(*$+0.05$*); BIQL is the best cell by* 0.15 *(*≈ 2 *sd) — the first replication of* BIQL>GCIQL *with positive guidance outside OGBench. On* `mw-pick-place` *headroom opened (*$H \approx 0.45$*) but guidance stayed flat (*$+0.05$ / -0.03*): per the decision tree this escalates pick-place to Ph. 2/3 (the goal channel and horizon defects are untouched by noise). Ph. 1 is confirmed as* a *cause (sufficient to unmask the ordering on push), not the whole story.*

Remark 4 (Post-registration results: P4.1–2 have run (2026-08-03))**.** ***P4.1 (bounded monolithic)****:* GCIQL *with* $\kappa \tanh(\cdot/\kappa)$*,* κ=75*, scores* 0.760 ± 0.037 *vs.* 0.732 ± 0.013 *unbounded — boundedness recovers at most* ~ 3 *of the* 19.5*-point gap. Per the decision tree, sub-claim (a) contributes little; completion (c) carries the phenomenon.* ***P4.2 (de-concentrated*** ψ***)****: removing the* ψ*-tower LayerNorm gives* 0.909 ± 0.032 *vs.* 0.927 ± 0.017 *— within noise of baseline, nowhere near the predicted drop toward* GCIQL*; sub-claim (b) is not load-bearing for performance. d=15:* 0.837 ± 0.027*, completing the saturation curve (*0.62/0.84/0.92 *at*

d=10/15/20*). Net verdict on Ph. 5: the conjunction survives but its weight shifts almost entirely onto (c) — bounded scale and uniform goal norms are incidental; the rank-restricted* completion *of the value family, grounded in the environment's own kernel spectrum (Remark 5), is the mechanism.*

Remark 5 (Post-registration: the grounding measurement (2026-08-03))**.** *The rank story is grounded in an intrinsic property of the environment. Graph-estimated ground truth (`eval/kernel_spectrum.py`: shortest paths over the validation data, temporal edges + one-hop-radius kNN shortcuts; no learning): the centered optimal-value matrix* $V = (\gamma^D - 1)/(1-\gamma)$ *has* 5–7 *dominant components carrying* 90% *of its energy (eff. rank* 5–11 *across estimator settings; top three components* $\approx$ *the block's three spatial coordinates). The* learned *value matrix* $\Phi\Psi^\top/\sqrt{d}$ *has the* ***same centered spectrum at both*** d=50 ***and*** d=512*: dims*$_{90}$ = 11/12*, dims*$_{95}$ = 16/18*, eff. rank* 10.1/10.6*, top components* 0.30/0.23/0.11 *— capacity-independent, unlike the embedding eff. rank (Remark 1). The invariant the d-sweep was looking for is the spectrum of the* expressed value function*, which matches the environment kernel and predicts the success saturation (*$d^\dagger \approx$ *dims*$_{95}$*:* 0.62/0.84/0.92 *at* d = 10/15/20*). Mechanistically this is the low-rank-MDP / successor-feature structure:* $\{V^*(\cdot, g)\}$ *spans a* $\approx$ 10*-dimensional function space because* d^* *factors through the 3-D block latent; the bilinear head is a spectral estimator of that family, and its off-distribution coherence (Remark 2) is inherited from the spectrum being a global object. Cross-env prediction:* $d^\dagger$ *tracks the goal-relevant latent dimension (cube-double* $\Rightarrow$ *roughly double), and near-rank-1 task families (single 3-D target) are the regime where monolithic critics suffice.*

5 Interpretation discipline

The five phenomena are deliberately stated so that the six experiments in Predictions 1–4 form a decision tree, not a shopping list:

Experiment	Outcome A $\Rightarrow$	Outcome B $\Rightarrow$
P1 shrink MW data	ranking appears: Ph. 1 confirmed	flat: escalate to Ph. 2
P2b eval-goal synthesis (MS)	guidance sign flips: Ph. 2	still negative: Ph. 3 (target broken)
P3b OGBench-style play on MS	cube ordering reappears: RQ1 closed	does not: simulator-intrinsic residual, profile dynamics
P4.1 bounded GCIQL	gap closes: boundedness is the asset	gap persists: completion (c) is the asset
P4.4 d-sweep	sharp at $d \approx 28$: rank story exact	gradual: rank is proxy, revisit

Priority order under the 12-hour budget: **P4.3** (hours; existing checkpoints, analysis only), **P4.1–2** (one config/one-file edit each, 500k cube runs), **P2b** (eval-side only, no retraining), then P1, P3. The first three alone are sufficient to either confirm or kill Ph. 5, which is the paper's central claim.

Honest caveats. (i) Ph. 2 explains the *sign* on MS and the FB asymmetry but not why MW-play-noise FLOWBC sometimes stays high; censoring (Ph. 1) covers that, and the two are confounded until Pred. 1 runs. (ii) The exploitation bound in Ph. 5(a) is an upper-bound argument, not an identity; the kill-test is P4.1, not the algebra. (iii) VIP rows are uninformative in the current matrix: the embedding collapsed to effective rank 1.0 (its P1 $\approx$ 0.04); fix its objective/normalization before citing it as a reference point.